



Foundations of Programming

Assignment No.: 14

PROBLEM STATEMENT: Write a C program to input a binary number from the user and convert it into a decimal number.

OBJECTIVE:

- To understand number system conversions.
- To learn how to convert binary numbers to decimal numbers in C.
- To practice the use of loops and arithmetic operations.
- To develop logical thinking for number system problems.

THEORY:

A binary number system uses only two digits: 0 and 1. Each digit of a binary number represents a power of 2 depending on its position. To convert a binary number into its decimal equivalent, each digit is multiplied by 2 raised to the power of its position and the results are added together. In C programming, this conversion can be implemented using loops and arithmetic operations.

PLATFORM: Linux

CODE:

```
#include<stdio.h>
```

```
#include<math.h>
```

```
int main()
```

```
{
```

```
    long long bin;
```

```
int dec=0, i=0, rem;

printf("Enter binary number: ");
scanf("%lld",&bin);

while(bin!=0)
{
    rem = bin%10;
    dec += rem * pow(2,i);
    bin = bin/10;
    i++;
}

printf("Decimal number = %d",dec);
}
```

INPUT: Binary number entered by the user.

OUTPUT: Displays the decimal equivalent of the given binary number.

SAMPLE INPUT:

```
Enter binary number: 1011
```

SAMPLE OUTPUT:

```
Decimal number = 11
```

CONCLUSION: Thus, the C program successfully converts a binary number entered by the user into its decimal equivalent using loops and arithmetic operations.

FAQs:

1. What is a binary number system?
2. How is a binary number converted into a decimal number?
3. What is the role of loops in binary to decimal conversion?
4. What is the role of the pow() function in this program?
5. What is the data type used to store binary and decimal numbers in this program?